18.100A

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These are my lecture notes from 18.100A, Real Analysis, at the Massachusetts Institute of Technology, taught this semester (Fall 2017) by Professor Arthur Mattuck[1].

I wrote these lecture notes in LaTeX in real time during lectures, so there may be errors and typos. I have lovingly pillaged Evan Chen's and Tony Zhang's formatting commands. Should you encounter an error in the notes, wish to suggest improvements, or alert me to a failure on my part to keep the web notes updated, please contact me at rmwu@mit.edu.

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1.1 Administrivia

- Professor David Jerison lives in office 2-272. Office hours are Tuesday 3-4p and Friday 3-4p in his office.
- Problem sets due Wednesdays in class, 2-190, every week excluding exam weeks.
- Homework policies: talk to each other, but don't copy others. Attribute peers and ideas. There's a single automatic extension of 1 week, but we must turn in every homework.
- This class references Baby Rudin. The book is "efficient" but lacks "informal guidance," so the professor is useful! However, it also lacks emphasis, so it's not clear what's important. Finally, the textbook also lacks pictures!
- The goal is to learn the foundations of calculus, infinite series, and differential equations. Furthermore, we should learn to read and write rigorous math.

"I mean I love probability theory, but not on my blackboard..."

1.2 Playing around with e

We begin with simple things.

> **Question 1.1.** What is e?

We have several answers from the class.

1. Euler said that
$$e = \lim_{n\to\infty} (1 + 1/n)^n.$$
2. Let $E(x)$ be the solution to $E'(x) = E(x)$, with $E(0) = 1$. Then $e \coloneqq E(1)$.
3. $e = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \dots$
4. Let $L(x)$ be the natural logarithmic equation, defined by Napier in 1600. Let $L(1) = 0, L'(1) = 1$ and $L(xy) = L(x) + L(y)$. Then e is the root of $L(e) = 1$.

In this course, we will learn how to choose among definitions, and we will see why they are all equivalent. For a long time, no one questioned those proofs. Then in 1800, Fourier invented his eponymous series. For instance,

$$\sum_{n=1}^{\infty} \frac{\sin nx}{n} = \frac{\pi}{2} - x, 0 < x < \pi$$

represents a sawtooth wave.

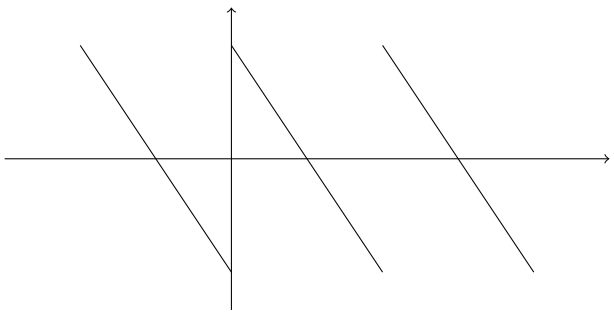


Figure 1: Sawtooth Fourier series.

At this point, people realized that the Fourier things would become really weird, and they'd never defined numbers to start with! From 1870 to 1920, lots of math happened. Weiner constructed the sequence

$$\sum_{n=1}^{\infty} G_n(w)\frac{\sin nx}{n}$$

where $G_n(w)$ are Gaussian random variables, for Brownian motion. The key is to look at the distance between functions. We begin with a simple attempt,

$$d(f,g) = \max_{0\le x\le T} |f(x) - g(x)|$$

Unfortunately, this distance function is not continuous. For Fourier series, the mean squared distance works better.

$$d(f,g) = \left(\int_0^{\pi} f(x) - g(x)dx\right)^2 \tag{1.1}$$

1.3 Real numbers

What are the real numbers?

Definition 1.2. A field F is a set with two operations called addition and multiplication that satisfies the following axioms.

1. The set is closed under addition. If $x, y \in F$, then $x + y \in F$.
2. Commutativity holds. $x + y = y + x$.
3. Associativity holds. $(x + y) + z = x + (y + z)$.
4. There exists an identity.
5. $\forall x \in F$, there exists an inverse $-x$ such that $x + (-x) = 0$.

The field satisfies the following "multiplication" axioms.

1. $xy \in F$.
2. $xy = yx$.
3. $(xy)z = x(yz)$.

4. $1x = x$.
5. $\forall x \neq 0$, there exists a $x^{-1} \in F$ such that $xx^{-1} = 1$.

Finally, the distributive property, $x(y + z) = xy + xz$.

tl;dr a field is an abelian group.

Example 1.3

TODO phrase this better Rationals m/n such that $m, n \in \mathbb{Z}$ and $n \neq 0$. The complex $\mathbb{C}$ with elements $a + bi$, where $a, b \in \mathbb{R}$ and $i^2 = -1$.

The integers $\mathbb{Z}$ are a non-example because there is no multiplicative inverse.

Definition 1.4. An **ordered set** S obeys two properties.

1. For every pair of elements $x, y \in S$, either $x < y$, $y < x$, or $x = y$. This property is known as a **trichotomy**.[2]
2. If $x < y$ and $y < z$, then $x < z$. This is **transitivity**.

Definition 1.5. An **ordered field** F is a field that is an ordered set satisfying

1. If $x > 0, y > 0$, then $x + y > 0$ and $xy > 0$.

[2] We use $x < y$ freely with $y > x$.

2 September 8, 2017

On Wednesday, they discussed limits rather than ordered fields. A sequence $\{x_n\}$ has a limit L if x_n is monotone and bounded. We use the notation

$$\lim_{n\to\infty} \{x_n\} = L. \tag{2.1}$$

2.1 Notation

2.1.1 Absolute values

We begin today by introducing notation and definitions. In baby calculus, we've defined absolute value to be

$$|a| :- \begin{cases} a & a > 0 \\ -a & a < 0 \\ 0 & a = 0 \end{cases} \tag{2.2}$$

However, this is not generalizable. Rather, we should define $|a|$ as either the distance away from 0 or the size of a. Absolute value has several properties.

1. $|ab| = |a| \cdot |b|$. This is trivial.
2. $|a + b| \leq |a| + |b|$. This is known as the triangle inequality.

Proof of triangle inequality. We know that $|a| \leq |a|$, which implies that $-|a| \leq a \leq |a|$. Similarly, we have that $-|b| \leq b \leq |b|$. We add the two inequalities

$$-(|a| + |b|) \leq a + b \leq |a| + |b|. \tag{2.3}$$

However, we can simplify this statement to

$$|a + b| \leq |a| + |b|, \tag{2.4}$$

as desired. □

> "Whenever you see three dots in an equation, it's the old-fashioned way of omitting induction."—apm

2.1.2 Approximation symbol

There are several in use, but the most common is the "double wiggle" $\approx$. In this class, the symbol $\underset{\epsilon}{\approx}$, where ϵ is a small real number, represents the distance between two numbers

$$a \underset{\epsilon}{\approx} b \coloneqq |a - b| < \epsilon. \tag{2.5}$$

Note that this is a strict inequality.

The transitive law states that $a \underset{\epsilon_1}{\approx} b, b \underset{\epsilon_2}{\approx} c$ implies that $a \underset{\epsilon_{1+2}}{\approx} c$.

Proof. We use the triangle equality.

$$\begin{aligned} |a - c| &= |(a - b) + (b - c)| \\ &\leq |a - b| + |b - c| \\ &< \epsilon_1 + \epsilon_2 \end{aligned}$$

□

2.1.3 Sufficiently large

The phrase "for $n \gg 1$" means "for n sufficiently large." So "$P(n)$ is true for $n \gg 1$" means that $\exists N \in \mathbb{R}$ such that $P(n)$ is true for all $n \geq N$.

Example 2.1

Prove that for $a > 1$, the sequence

$$x_n = \frac{a^n}{n}$$

is monotone for $n \gg 1$.

Proof. We show that $x_{n+1}/x_n > 1$ for some N.

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{a^{n+1}}{n+1} \cdot \frac{n}{a^n} \\ &= \frac{an}{n+1} > 1 \end{aligned}$$

We solve the inequality to find that $n > 1/(1-a)$.

□

2.2 Limits

Recall the definition,

$$\lim_{n\to\infty} \{x_n\} = L.$$

Given $\epsilon > 0$, $x_n \underset{\epsilon}{\approx} L$ for $n \gg 1$ where the sufficiently large N depends on ϵ. We also use the prefixes "for any" or "for all."